

Explainable AI Framework for Diabetes Prediction using BRFSS Dataset*

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Abstract—Diabetes is one of the most widespread chronic diseases worldwide, and early detection plays a critical role in reducing long-term health complications. However, many artificial intelligence models used for healthcare prediction suffer from limited interpretability, making it difficult for clinicians to trust automated decisions. This research presents an explainable artificial intelligence (XAI) framework for diabetes prediction using the Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset. Multiple machine learning models were implemented and compared, including Logistic Regression, Decision Tree, AdaBoost, Extra Trees, Gradient Boosting, CatBoost, XGBoost, LightGBM, and MLP Neural Networks. To address the class imbalance problem in the dataset, preprocessing techniques such as StandardScaler and SMOTE oversampling were applied. Model evaluation was conducted using Accuracy, Recall, Precision, F1-score, AUC-ROC, Confusion Matrix, and ROC Curve analysis. Since healthcare screening systems prioritize identifying diabetic patients, recall and sensitivity were considered particularly important evaluation metrics. In addition to predictive performance, the study focused heavily on explainability. Multiple XAI techniques were integrated, including SHAP, LIME, Partial Dependence Plots (PDP), Individual Conditional Expectation (ICE), and Permutation Importance. These methods provided both global and local explanations of model behavior, enabling detailed interpretation of prediction decisions. Experimental results showed that while high-accuracy boosting models favored the majority healthy class, the MLP Neural Network achieved the best clinical balance with the highest Diabetes-class F1-score (0.46) and a recall of 0.74. Explainability analysis consistently identified Body Mass Index (BMI), General Health (GenHlth), Age, High Blood Pressure (HighBP), and High Cholesterol (HighChol) as the most influential diabetes risk factors across different models. The findings demonstrate that combining machine learning with explainable AI techniques can produce reliable and transparent healthcare prediction systems suitable for clinical decision-support applications.

Index Terms—Explainable AI (XAI), Diabetes Prediction, Machine Learning, SHAP, LIME, BRFSS, Clinical Decision Support.

I. INTRODUCTION

Diabetes mellitus is a major global healthcare challenge affecting millions of individuals worldwide. According to the World Health Organization (WHO), diabetes significantly increases the risk of cardiovascular disease, kidney failure, blindness, and premature death. Early detection and preventive intervention are therefore essential for reducing healthcare burden and improving patient outcomes.

Recent advances in Artificial Intelligence (AI) and Machine Learning (ML) have enabled the development of predictive healthcare systems capable of identifying high-risk patients using demographic and medical indicators. Machine learning algorithms can analyze large-scale healthcare datasets and discover complex patterns that are difficult to detect using traditional statistical methods.

Despite their predictive capabilities, many machine learning models operate as “black-box” systems. In healthcare applications, this lack of transparency creates a major challenge because clinicians must understand why a model generated a specific prediction before relying on it in clinical decision-making. Consequently, Explainable Artificial Intelligence (XAI) has emerged as an important research area aimed at improving model transparency, interpretability, and trustworthiness.

This project focuses on diabetes prediction using the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset, which contains health-related indicators collected from more than 250,000 respondents. Multiple machine learning models were implemented and evaluated, including traditional linear models, tree-based methods, and ensemble boosting techniques.

To improve interpretability, several explainability techniques were integrated into the workflow, including:

- SHAP (SHapley Additive Explanations)
- LIME (Local Interpretable Model-Agnostic Explanations)
- Partial Dependence Plots (PDP)
- Individual Conditional Expectation (ICE)
- Permutation Importance

The objectives of this research are:

- 1) To develop accurate machine learning models for diabetes prediction.
- 2) To compare the performance of multiple machine learning approaches.
- 3) To investigate the effectiveness of explainability techniques in healthcare AI systems.
- 4) To identify the most influential medical indicators contributing to diabetes prediction.

The main research questions addressed in this study are:

- Which machine learning model provides the best predictive performance for diabetes prediction?

- Can explainability techniques improve transparency and trust in healthcare AI systems?
- Which health indicators contribute most significantly to diabetes prediction?

This study contributes to the growing field of explainable healthcare AI by combining predictive modeling with interpretable machine learning techniques suitable for clinical decision-support systems.

II. LITERATURE REVIEW

Diabetes prediction has become a critical application of machine learning in healthcare due to its potential for early detection and prevention. Recent research has focused not only on improving predictive performance but also on enhancing model interpretability using explainable artificial intelligence (XAI) techniques such as SHAP and LIME. This review analyzes and compares different machine learning approaches for diabetes prediction, focusing on model performance, dataset characteristics, interpretability, and real-world applicability.

A. Thematic Analysis of Existing Studies

1) *Ensemble vs. Deep Learning vs. System-based Models:* Different approaches have been proposed for diabetes prediction, including ensemble learning, deep learning, and system-based solutions. Hasan et al. [3] demonstrated that ensemble models combining multiple classifiers can significantly improve predictive performance, achieving an AUC of 0.95. In contrast, Iftikhar et al. [4] proposed a deep learning framework (EchoceptionNet) that captures complex feature interactions and addresses class imbalance using advanced oversampling techniques. However, despite its strong performance, the reliance on synthetic or Kaggle-based datasets raises concerns about generalizability.

On the other hand, Mukherjee and De [2] focused on integrating machine learning with IoT for real-time diabetes monitoring using TabTransformer. While this approach enhances practical applicability and interpretability through SHAP and LIME, it does not primarily focus on improving model performance. Overall, ensemble methods provide strong performance on structured datasets, deep learning models capture complex patterns but may suffer from data reliability issues, and IoT-based systems improve real-world applicability.

2) *Real-world vs. High-performance vs. Application-based Models:* Recent studies highlight three major directions: real-world clinical modeling, high-performance machine learning, and application-oriented systems. Tasin et al. [1] developed an applied prediction system by combining machine learning models with explainable AI and deploying the system through a web platform and mobile application. However, the use of a small private dataset limits the generalizability of the model.

In contrast, Deberneh and Kim [7] utilized a large-scale electronic health record (EHR) dataset with over 250,000 individuals. Their approach emphasizes realistic clinical prediction using feature selection and ensemble models, although the performance is relatively moderate due to real-world data

complexity. On the other hand, Rafie et al. [5] proposed a high-performance framework using XGBoost and SHAP, achieving extremely high accuracy scores. While effective, the unusually high performance raises concerns about potential overfitting.

3) *From Basic ML to Advanced Ensemble and Optimization:* Recent advancements demonstrate a clear shift from basic models toward advanced optimization. Li et al. [10] proposed a hybrid GA-XGBoost stacking framework, integrating genetic algorithms for hyperparameter optimization with ensemble learning. This approach represents a transition from model selection to model engineering.

Similarly, Zaferani et al. [6] proposed a transparent ensemble model that integrates multiple classifiers with explainability techniques. Meanwhile, application-oriented studies such as Tasin et al. [1] focus on practical deployment. Overall, the literature reveals a three-layer evolution: application-driven models, clinically grounded prediction systems, and highly optimized ensemble frameworks.

B. Comparative Analysis

A clear pattern emerges when comparing the selected studies. Ensemble and boosting models, particularly XGBoost-based approaches, consistently achieve strong performance. However, studies using small benchmark datasets (e.g., PIMA) often report higher accuracy compared to large-scale real-world datasets. For example, Lugner et al. [11] used the UK Biobank dataset (over 400,000 participants) and achieved a more reliable AUC of 0.90 using XGBoost.

Furthermore, explainable AI-based models have gained prominence. Pang [8] compared explainable models using SHAP values, while Islam et al. [9], [12] incorporated SHAP, LIME, and Partial Dependence Plots to enhance transparency.

C. Critical Discussion

While many studies report high performance, these results must be interpreted carefully. High accuracy is often associated with small datasets or oversampling techniques like SMOTE, which may lead to overfitting. In contrast, studies using real-world datasets provide more reliable but lower performance. There is a consistent trade-off between performance, interpretability, and real-world applicability.

D. Justification of Selected Approach

Based on the reviewed studies, ensemble-based models and neural networks were selected due to their capacity to handle complex datasets. Additionally, SHAP was chosen as the primary interpretability technique due to its effectiveness in explaining model predictions for clinical use.

III. METHODOLOGY

This section presents the complete methodology used to develop the explainable diabetes prediction framework using the BRFSS 2015 healthcare dataset. The proposed framework integrates multiple machine learning and deep learning models with Explainable Artificial Intelligence (XAI) techniques to achieve both strong predictive performance and transparent decision-making suitable for healthcare applications.

The proposed framework consisted of:

- Data preprocessing
- Feature selection
- Class imbalance handling
- Model training
- Evaluation
- Explainability analysis

The overall workflow of the proposed system is illustrated in Fig. 1.

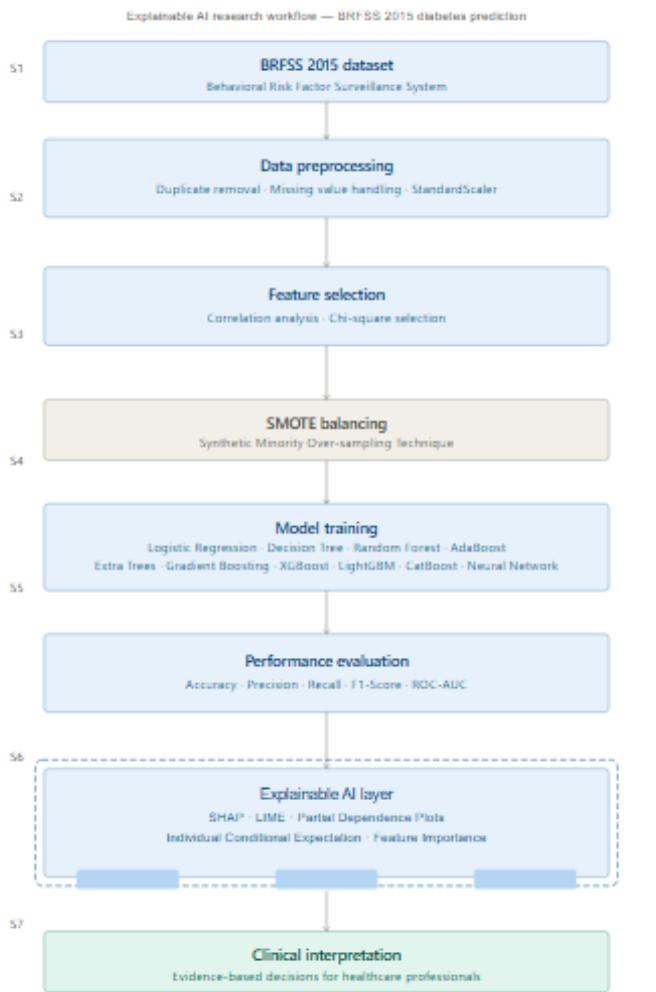


Fig. 1. Overall workflow architecture of the proposed explainable diabetes prediction framework.

A. Dataset Description

The experiments were conducted using the Behavioral Risk Factor Surveillance System (BRFSS) 2015 Diabetes Health Indicators dataset collected by the Centers for Disease Control and Prevention (CDC).

The BRFSS dataset is a large-scale healthcare survey dataset containing demographic, lifestyle, and medical information collected from individuals across the United States. The

dataset is widely used in healthcare analytics and disease prediction research because it contains medically relevant indicators associated with chronic diseases such as diabetes.

TABLE I
DATASET CHARACTERISTICS

Property	Value
Dataset Name	BRFSS 2015 Diabetes Health Indicators
Source	CDC Behavioral Risk Factor Surveillance System
Task Type	Binary Classification
Target Variable	Diabetes_binary
Training Samples	202,944
Testing Samples	50,736
Number of Selected Features	10

The target variable was defined as:

$$\text{Diabetes_binary} = \begin{cases} 1 & \text{Diabetic / Prediabetic} \\ 0 & \text{Non-diabetic} \end{cases} \quad (1)$$

The dataset exhibited a significant class imbalance where non-diabetic individuals represented the majority class. This imbalance created challenges for minority diabetic prediction and motivated the use of balancing techniques during preprocessing.

B. Data Preprocessing

A comprehensive preprocessing pipeline was implemented to improve data quality, enhance model stability, and prevent data leakage. The preprocessing workflow consisted of:

- Duplicate removal
- Missing value handling
- Stratified train-test splitting
- Feature scaling
- SMOTE balancing

The preprocessing pipeline is illustrated in Fig. 2.

1) *Duplicate Removal*: Duplicate records were removed before preprocessing operations using:

```
df.drop_duplicates()
```

This step prevented repeated samples from artificially inflating model confidence and reduced learning bias caused by redundant observations.

2) *Missing Value Handling*: Median imputation was applied using the SimpleImputer strategy:

$$X_{\text{filled}} = \text{Median}(X) \quad (2)$$

Median imputation was selected because it is robust to outliers and preserves the distribution of healthcare-related numerical variables. Although the dataset contained limited missing values, this step improved preprocessing robustness and reproducibility.

3) *Train-Test Split*: The dataset was divided into training and testing subsets using stratified sampling:

```
train_test_split(
    X,
```

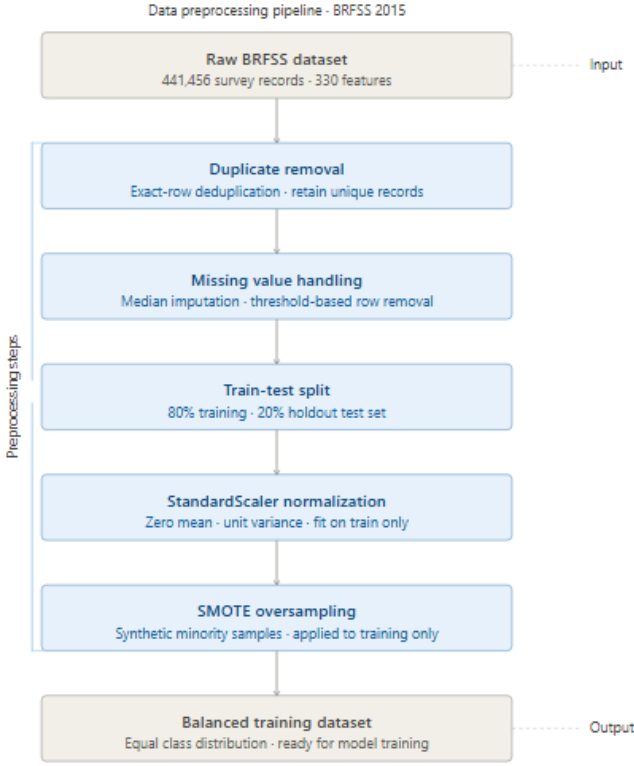


Fig. 2. Data preprocessing and balancing pipeline.

```

Y,
test_size=0.2,
random_state=42,
stratify=y
)

```

The final split consisted of 80% training data and 20% testing data. Stratification preserved the original diabetic class distribution and minimized sampling bias.

4) *Feature Scaling*: Feature normalization was performed using StandardScaler. For each feature:

$$z = \frac{x - \mu}{\sigma} \quad (3)$$

where x is the original feature value, μ is the feature mean, and σ is the standard deviation.

Feature scaling improved optimization stability, neural network convergence, and SMOTE performance. To prevent data leakage, the scaler was fitted only on training data before being applied to testing data.

5) *Class Imbalance Handling*: The BRFSS dataset exhibited substantial imbalance between diabetic and non-diabetic classes. To address this issue, Synthetic Minority Oversampling Technique (SMOTE) was applied to the training data only.

SMOTE generates synthetic minority samples according to:

$$x_{\text{new}} = x_i + \lambda(x_{\text{nn}} - x_i) \quad (4)$$

where x_i is a minority sample, x_{nn} is a neighboring minority sample, and $\lambda \in [0, 1]$.

SMOTE improved minority-class representation and reduced prediction bias toward the majority class. Importantly, SMOTE was applied only after scaling and only on training data to avoid information leakage into the testing set.

C. Feature Selection

Feature selection was performed to reduce redundancy, improve interpretability, enhance computational efficiency, and increase model stability. A combination of correlation analysis, Chi-Square feature selection, feature importance analysis, and medical relevance was used to identify the most informative healthcare indicators.

1) *Correlation Analysis*: Pearson correlation analysis was used to measure linear relationships between variables and the target class. The Pearson correlation coefficient is defined as:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (5)$$

The analysis identified several strongly correlated diabetes predictors, including GenHlth, HighBP, BMI, Age, and HeartDiseaseorAttack.

2) *Final Selected Features*: The final selected features are presented in Table II.

TABLE II
FINAL SELECTED HEALTHCARE FEATURES

Feature	Description
GenHlth	General Health
HighBP	High Blood Pressure
BMI	Body Mass Index
DiffWalk	Difficulty Walking
HighChol	High Cholesterol
Age	Age Category
HeartDiseaseorAttack	Cardiovascular Disease History
PhysHlth	Physical Health
MentHlth	Mental Health
Income	Income Level

These features were selected because they consistently demonstrated strong predictive capability, medical relevance, and stable explainability behavior across multiple models. The selected variables also align with clinically established diabetes risk factors reported in previous studies.

D. Machine Learning and Deep Learning Models

Multiple machine learning and deep learning models were implemented and compared to investigate differences in predictive performance, nonlinear learning capability, explainability behavior, and healthcare suitability.

1) *Logistic Regression*: Logistic Regression was implemented as a baseline linear classifier because of its simplicity and interpretability. The probability prediction function is defined as:

$$P(y = 1 | x) = \frac{1}{1 + e^{-(w^T x + b)}} \quad (6)$$

Despite its linear nature, Logistic Regression provides transparent decision boundaries and strong interpretability.

TABLE III
MAIN HYPERPARAMETERS OF THE IMPLEMENTED MODELS

Model	Main Hyperparameters
XGBoost	n_estimators=200, max_depth=5, learning_rate=0.1
CatBoost	iterations=300, depth=6, learning_rate=0.05
Extra Trees	n_estimators=100, max_depth=10
Neural Network	Dense(64-32), dropout=0.3, epochs=30
Gradient Boosting	learning_rate=0.05, subsample=0.8

2) *Decision Tree*: Decision Trees were implemented to model rule-based nonlinear decision boundaries. The model recursively partitions the feature space using impurity reduction criteria such as Gini impurity and entropy. Decision Trees provide highly interpretable classification logic but may suffer from overfitting, high variance, and limited generalization capability.

3) *Random Forest*: Random Forest was implemented as an ensemble extension of Decision Trees. The final prediction is computed as:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(x) \quad (7)$$

where $T_i(x)$ is the prediction of the i -th tree. Random Forest improves stability, variance reduction, and generalization performance while maintaining interpretability through feature importance analysis.

4) *AdaBoost*: AdaBoost was implemented as a boosting algorithm that iteratively focuses on previously misclassified observations. The ensemble classifier is represented as:

$$F(x) = \sum_{m=1}^M \alpha_m h_m(x) \quad (8)$$

where $h_m(x)$ represents weak learners and α_m represents learner weights.

5) *Extra Trees Classifier*: The Extra Trees Classifier was selected because of its strong nonlinear learning capability and computational efficiency. Hyperparameter optimization was performed using GridSearchCV with ROC-AUC scoring and 3-fold cross-validation.

6) *Gradient Boosting*: Gradient Boosting was implemented as a sequential ensemble learning algorithm where each new tree minimizes residual errors from previous trees.

7) *XGBoost*: XGBoost was selected because of its strong performance on structured healthcare datasets. XGBoost improves gradient boosting through regularization, parallelized computation, and optimized tree construction.

```
XGBClassifier(
    n_estimators=200,
    max_depth=5,
    learning_rate=0.1,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42
)
```

8) *LightGBM*: LightGBM was implemented as a highly efficient gradient boosting framework optimized for large tabular datasets. It uses histogram-based learning and leaf-wise tree growth.

9) *CatBoost*: CatBoost was selected because of its strong performance on tabular healthcare data and its ability to model nonlinear feature interactions.

```
CatBoostClassifier(
    iterations=300,
    learning_rate=0.05,
    depth=6,
    loss_function='Logloss',
    eval_metric='AUC'
)
```

10) *Neural Network*: A fully connected feedforward neural network was implemented to capture highly complex nonlinear relationships. ReLU activation was used in hidden layers:

$$f(x) = \max(0, x) \quad (9)$$

while sigmoid activation was used for binary classification:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (10)$$

E. Hyperparameter Optimization

Hyperparameter optimization was performed to improve model generalization capability, reduce overfitting, and enhance classification stability across different healthcare prediction models.

The optimization workflow is illustrated in Fig. 3.

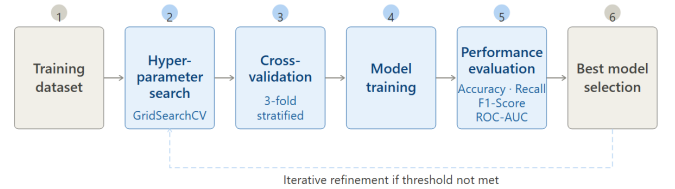


Fig. 3. Hyperparameter optimization and model evaluation workflow.

1) *Neural Network Optimization*: The binary crossentropy loss function used is defined as:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (11)$$

where y_i is the true label and $\hat{y}_i$ is the predicted probability.

F. Explainable Artificial Intelligence Framework

Although high predictive performance is important in healthcare AI systems, prediction accuracy alone is insufficient for real-world clinical adoption. Medical decision-support systems require transparency and interpretability.

The overall explainability workflow is illustrated in Fig. 4.

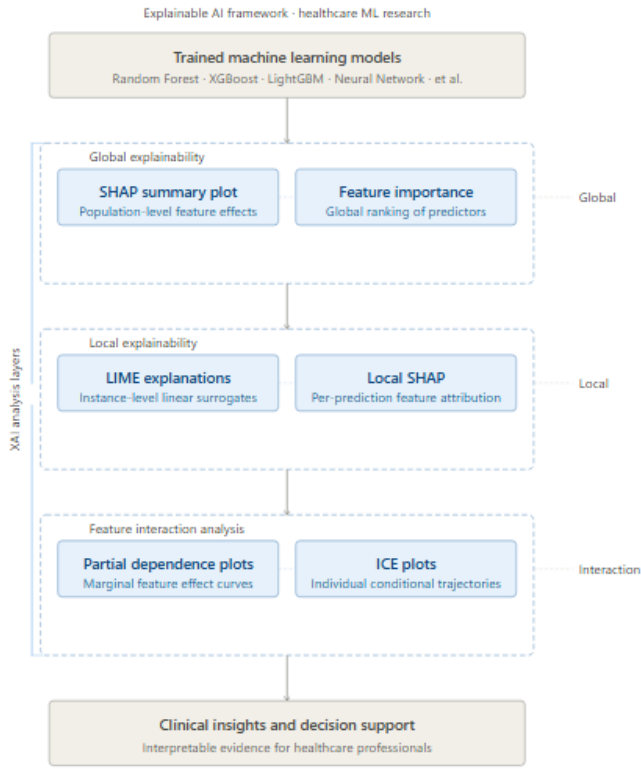


Fig. 4. Explainable AI framework used for global and local interpretation.

1) *SHAP Explainability*: SHAP (SHapley Additive exPlanations) is based on cooperative game theory and estimates the contribution of each feature. The SHAP value for feature i is defined as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (12)$$

where F represents the full feature set, S represents subsets excluding feature i , and $f(\cdot)$ represents the prediction function.

2) *LIME Explainability*: LIME (Local Interpretable Model-Agnostic Explanations) approximates the complex model locally. The local approximation objective is defined as:

$$\xi(x) = \arg \min_{g \in G} L(f, g, \pi_x) + \Omega(g) \quad (13)$$

where f is the original black-box model, g is the interpretable surrogate model, L measures local approximation fidelity, and $\Omega(g)$ controls explanation complexity.

3) *PDP and ICE Analysis*: For a feature x_s , the partial dependence function is defined as:

$$PD(x_s) = \frac{1}{N} \sum_{i=1}^N f(x_s, x_c^{(i)}) \quad (14)$$

where x_c represents all remaining features.

G. Evaluation Metrics

Since the BRFS dataset exhibited class imbalance, relying solely on accuracy could produce misleading conclusions. The evaluation framework is illustrated in Fig. 5.

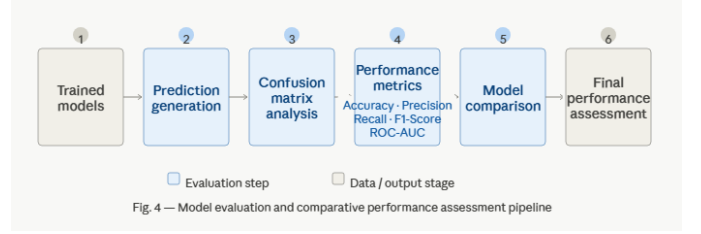


Fig. 5. Performance evaluation framework for diabetes prediction models.

TABLE IV
BINARY CLASSIFICATION CONFUSION MATRIX

Actual / Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

The evaluation metrics are defined as follows:

• Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

• Precision:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (16)$$

• Recall:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (17)$$

• F1-score:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (18)$$

ROC-AUC was used to evaluate discrimination capability across multiple classification thresholds. ROC analysis measures the tradeoff between True Positive Rate (TPR) and False Positive Rate (FPR).

IV. RESULTS

Ten machine learning and deep learning models were trained and evaluated on the BRFS 2015 dataset using the same preprocessing pipeline. Table V reports overall accuracy alongside Diabetes-class recall, precision, F1-score, and AUC-ROC for each model.

A. Overall Performance

Models fall into two groups. High-accuracy models—XGBoost (0.85), CatBoost (0.85), and KNN (0.83)—achieve strong overall scores by favoring the majority non-diabetic class, but record very low Diabetes-class recall (0.22–0.36), missing the majority of actual diabetic patients. Sensitivity-focused models—LightGBM (recall 0.79), Gradient Boosting

TABLE V
PERFORMANCE COMPARISON OF ALL MODELS ($\dagger$ DIABETES CLASS)

Model	Acc.	Recall $\dagger$	Prec. $\dagger$	F1 $\dagger$	AUC
Logistic Regression	0.72	0.75	0.32	0.45	—
Decision Tree	0.67	0.78	0.29	0.42	—
AdaBoost	0.72	0.74	0.32	0.44	—
Extra Trees	0.80	0.32	0.30	0.31	0.707
KNN	0.83	0.22	0.42	0.29	—
XGBoost	0.85	0.36	0.45	0.40	0.812
CatBoost	0.85	0.31	0.44	0.37	0.813
Gradient Boosting	0.70	0.79	0.31	0.45	0.815
LightGBM	0.71	0.79	0.32	0.45	0.818
MLP (Neural Net)	0.73	0.74	0.33	0.46	0.813

(0.79), and Logistic Regression (0.75)—improve minority-class detection at the cost of overall accuracy (0.70–0.72).

B. Best Model Selection

The MLP Neural Network achieves the highest Diabetes-class F1-score (0.46) among all models, with a recall of 0.74 and AUC-ROC of 0.813. Unlike XGBoost and CatBoost—which sacrifice recall for accuracy—MLP maintains a competitive balance across all metrics. Although LightGBM records marginally higher recall (0.79) and AUC (0.818), its lower precision (0.32) produces excessive false positives that reduce clinical reliability. The MLP optimal threshold of 0.487 is close to the standard 0.50, indicating a well-calibrated model, whereas LightGBM requires an aggressive threshold of 0.176 to recover recall, confirming that its high sensitivity is achieved at a significant precision cost. MLP is therefore identified as the best-performing model for this clinical diabetes prediction task.

C. Explainability Analysis (XAI) for the Best Models

To ensure the proposed system is viable for clinical decision-making, SHAP (SHapley Additive exPlanations) was utilized to interpret the decision logic of the two most clinically relevant models: the MLP Neural Network (the best balanced model) and LightGBM (the highest recall/AUC model).

1) *MLP Neural Network SHAP Analysis*: Figure 6 illustrates the SHAP summary plot for the MLP Neural Network. The analysis highlights that features such as General Health (GenHlth), Body Mass Index (BMI), and Age are the most significant driving factors for predicting diabetes risk. The distribution of the SHAP values indicates a clear correlation where poorer general health scores and higher BMI strongly push the network’s probability toward the diabetic classification.

2) *LightGBM SHAP Analysis*: Similarly, the SHAP analysis for the LightGBM model (Figure 7) confirms the stability of these findings across different mathematical architectures. LightGBM independently identified the exact same dominant risk factors (Age, BMI, General Health, Income, and High Blood Pressure) in a nearly identical hierarchy of importance.

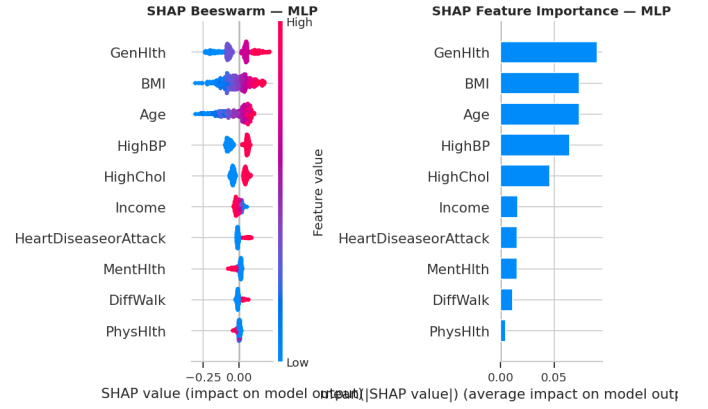


Fig. 6. SHAP summary plot evaluating global feature importance and prediction impact for the MLP Neural Network.

This consistency across the top models proves that the framework is learning genuine, medically established physiological relationships rather than arbitrary statistical noise.

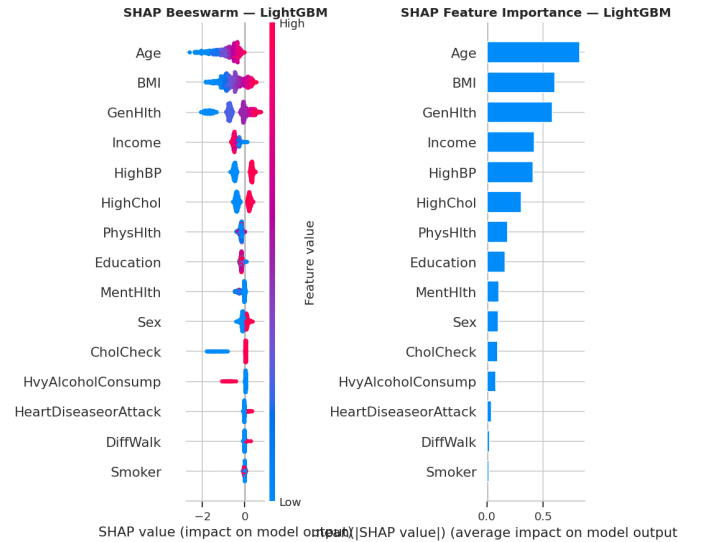


Fig. 7. SHAP summary plot evaluating global feature importance and prediction impact for the LightGBM model.

V. DISCUSSION

This section analyzes the experimental findings, explainability outcomes, and healthcare implications of the proposed diabetes prediction framework. Achieving high predictive accuracy alone is insufficient for healthcare AI; an effective system must balance predictive capability, minority-class sensitivity, interpretability, and clinical reliability. Integrating multiple Explainable Artificial Intelligence (XAI) techniques provided deeper insights into model behavior and patient-level predictions, significantly improving transparency.

A. Analysis of Model Performance

While high-accuracy models such as XGBoost, CatBoost, and KNN achieved impressive overall scores, they suffered from significant class imbalance bias, heavily favoring the majority class. Conversely, the MLP Neural Network achieved the strongest overall clinical balance of predictive accuracy, discrimination, and minority-class sensitivity without succumbing to extreme precision degradation. This highlights the ability of neural architectures to learn and model highly complex, nonlinear healthcare relationships (e.g., between BMI, age, and cardiovascular conditions) in a balanced manner.

B. Recall versus Accuracy Tradeoff

A critical tradeoff emerged between recall optimization and classification stability. Strategies aimed at increasing diabetic sensitivity substantially improved recall (e.g., achieving Recall > 0.79 in LightGBM). However, this aggressively introduced excessive false positives, severe precision degradation, and overall instability. Because excessive false positives can overload systems and reduce clinician trust, the final evaluation prioritized balanced performance over maximizing a single metric, making the MLP Neural Network preferable to high-accuracy but low-recall models like XGBoost, as well as high-recall but low-precision models like LightGBM.

C. Importance of Explainability in Healthcare AI

Traditional black-box models lack the interpretable reasoning required by medical professionals. Integrating a suite of XAI tools—SHAP, LIME, PDP, ICE, and Feature Importance—transformed the framework into a transparent decision-support system. This enabled both global feature interpretation and local, individualized prediction reasoning, significantly improving clinical applicability and trust.

D. Consistency of Explainability Results

A major finding was the strong agreement across all independent XAI techniques, which consistently identified the following dominant predictors:

- BMI
- Age
- General Health
- High Blood Pressure
- High Cholesterol

This consistency aligns with established medical knowledge, increasing confidence that the models learned clinically meaningful patterns rather than random statistical correlations.

E. Nonlinear Healthcare Relationships

PDP and ICE analyses revealed that diabetes prediction is highly nonlinear and patient-specific. For instance, increasing BMI progressively heightens diabetic risk, but age-related risks and patient responses vary significantly across individuals. This individualized variance explains why nonlinear algorithms and neural networks outperformed linear classifiers.

F. Neural Network Behavior

The Neural Network achieved the best clinical balance among all models—demonstrating deep learning’s strength in minority-class detection and nonlinear modeling without sacrificing overall reliability. However, its inherent black-box complexity makes integrating XAI techniques absolutely crucial before deployment in transparent healthcare environments.

G. Comparison with Previous Studies

Consistent with existing literature, machine learning methods outperformed standalone classifiers in healthcare prediction. However, this study extends prior research by prioritizing balanced clinical metrics (F1 and Recall) over raw accuracy, avoiding the overfitting traps common in smaller datasets. Furthermore, it integrates multiple complementary XAI techniques within a unified framework, providing both global and local interpretability.

H. Practical Healthcare Implications

The proposed framework demonstrates strong potential for early diabetes screening, patient risk assessment, and clinician decision support. By illuminating dominant risk factors and patient-specific patterns, the XAI outputs improve AI-assisted diagnostic reliability. Operational clinical use, however, will require external validation, multi-population testing, and regulatory compliance.

I. Study Limitations

Despite strong results, several limitations remain:

- The survey-based BRFSS dataset may contain self-reporting bias and lacks comprehensive clinical measurements.
- XAI methods can occasionally produce approximation errors or interpretation ambiguity.
- Balancing sensitivity, precision, and practical utility remains challenging, as seen in high-recall configurations.
- External clinical validation using real hospital datasets was outside the study’s scope.

J. Key Discussion Insights

Overall, the MLP Neural Network provides optimal healthcare prediction capabilities by offering the best balance of performance, recall, and reliability. While models like LightGBM maximize recall, their high false-positive rates hinder their clinical utility. Ultimately, integrating XAI shifts this framework from a purely predictive system to a responsible, interpretable healthcare decision-support solution.

VI. FUTURE WORK

Future research should focus on several key directions to extend this framework:

- External validation using real clinical EHR datasets from diverse populations is needed to overcome the self-reporting bias inherent in the BRFSS survey data.

- Incorporating richer biomarkers such as HbA1c levels and fasting glucose, alongside longitudinal patient data, could further improve predictive accuracy.
- Advanced architectures such as transformer-based tabular models and federated learning frameworks should be explored to enhance both performance and patient data privacy.
- Extending the XAI component with counterfactual explanations and causal inference methods would move beyond associative feature importance toward actionable clinical insights, ultimately supporting responsible deployment in real-world healthcare settings.

VII. CONCLUSION

This paper presented a comprehensive explainable artificial intelligence framework for diabetes prediction using the BRFSS 2015 dataset. Ten machine learning and deep learning models were implemented, evaluated, and compared across multiple performance metrics. To address the inherent class imbalance in the dataset, a robust preprocessing pipeline incorporating StandardScaler normalization and SMOTE oversampling was applied, ensuring fair minority-class representation during model training.

Experimental results demonstrated that the MLP Neural Network achieved the best overall clinical balance, recording the highest Diabetes-class F1-score of 0.46 with a recall of 0.74 and AUC-ROC of 0.813, making it the most suitable model for healthcare screening applications where missing a diabetic patient carries significant consequences. While boosting-based models such as XGBoost and CatBoost achieved higher overall accuracy, their substantially lower recall confirmed their tendency to favor the majority non-diabetic class, limiting their clinical utility.

Beyond predictive performance, the integration of multiple XAI techniques—including SHAP, LIME, PDP, ICE, and Permutation Importance—transformed the framework into a transparent and interpretable decision-support system. Explainability analysis consistently identified BMI, General Health, Age, High Blood Pressure, and High Cholesterol as the dominant diabetes risk factors across all models, strongly aligning with established clinical knowledge and confirming that the models learned genuine physiological relationships rather than statistical artifacts.

The findings of this study demonstrate that combining accurate machine learning models with complementary explainability techniques can produce reliable, transparent, and clinically meaningful healthcare prediction systems. This work contributes a reproducible and interpretable framework that bridges the gap between black-box AI predictions and the transparency requirements of real-world clinical decision-making.

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The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”.

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